

ActInf Livestream #032.2 ~ "Stochastic Chaos and Markov Blankets"

Livestream | 2:17:04

[Music]

[Music]

[Music]

hello everyone welcome to actin flab to
act in flab live stream number 32.2 it's
november 16 2021

welcome to the active lab

we are a participatory online lab that
is communicating learning and practicing
applied active inference you can find us
at some of the links here on this slide
this is recorded and an archived live
stream so please provide us with
feedback so we can improve on our work
all backgrounds and perspectives are
welcome here and we'll be following good
and fun video etiquette for live streams
at this short link you can see some of
the upcoming streams this is our second
participatory group discussion on the
stochastic chaos and markov blankets
paper

and then in the second two weeks of
november we're going to be discussing
thinking like a state by our friend and
co-participant aval and we haven't yet
set the papers for december so if you
know an author who you'd like to invite
or if you yourself would like to join
for several weeks of discussion of one
of your works then just let us know
today in active stream number 32.2 we're
aiming to learn and discuss and ask
what if and what is and hold all these
different questions at once

we're in the second participatory
discussion around the paper stochastic
chaos and markov blankets by fristen
heinz ultzhoffer decosta and par
and last week we had an awesome
discussion in the dot one
and we uh got right up to the
cliffhanger where the the m word
actually was deployed and i think today
we're gonna um sprint up to that cliff
and then jump off so it'll be a fun
discussion for those who are watching
live

totally right questions and comments in
the live chat and stephen and dean and i
and anyone else who joins will be just
more than happy to
hear what you have to share and to um
[Music]

roll with it as you write things so
let's begin just by giving a
introduction or warm-up so
we can say hello and then also just
maybe what was exciting about the paper
overall or since we all participated in
the dot one

what are you looking forward to in the
dot 2

or maybe how have you changed how you
thought about it over the previous week
so i'm daniel i'm a researcher in
california

and
i'll save what i'm excited for until a
little bit later and i'll pass it to
dean

hey good morning uh my name is dean i'm
in calgary uh in the

in the one weeks time since we
left off i

i think this paper more than many of the
other people we've had conversations
around breathes life into something that
is typically perceived as
not very

alive it's just it's just a sort of a
point to and an explanation of but i
think this is one of those
papers where

when people pick it up and start having
conversations with one another about
the kind of work that's being done here
it makes it really really interesting
doesn't give us necessarily
the answers that the authors
could project would be given and i
really like that

thank you dean stephen i'll pass it to
stephen

good morning as well from toronto um
i'm really interested in this paper i
think because it

it does um give a unifying
perspective it does help with that
unifying perspective even though at
first when you see things like
stochastic chaos it can be like oh we're
going down another rabbit hole etcetera
etcetera

um but actually
rather than always justifying or talking
about applied active influence influence
through applications and quotient
applications there is some benefit to
this kind of coming to the foundation
and so i'm excited by that and also

actually following the conversation
we're having earlier in the fact that we
have this um complexity weekend
is some of the ways that complexity can
be talked about outside of a more
woo-woo kind of
conversational sense at these
foundational levels which again seems to
get into unnecessary
um detail
ultimately i think it starts to allow
some of these crossovers between
applications of complexity and this to
happen which for me personally i don't
think i was necessarily able to
articulate a year ago so i think
some of this work
does just help fill in some of those
gaps
thanks i agree like the system that's
picked up and turned around and
remodeled throughout the course of the
paper in a very
narrative driven way is the lorenz
attractor which is like the classic
complex systems model
so it just kind of
plays into
contemporary work on complexity and
so many other areas
but and then also dean i i really like
what you said like
the authors i hope will be able to join
for a future presentation so we're in
contact they probably won't join today
but
what the authors are putting out there
and then how that's received and

communicated around and explored
that's what's fun and um
i know that we'll have many
cool things to explore today
so let's
jump to the big questions
then we'll look at the road map and try
to remember where we got to yesterday
and maybe a different big question is
striking either of you or anyone in the
live chat today and then we'll
try to remember how far on our road trip
we got
last tuesday and then let's see which
sections we're going to get to and we
have a bunch of other things kind of
written down so the big questions that
we had previously asked were what is a
good model of thingness in a chaotic
dynamic anticipative world
how might this model or approach to
thingness speak to system sentience
and then there in little tiny
font it's a sentence not in the
phenomenological
like experiential
but greeting cervel good to see you
but sentience in the set in the sense of
feeling thinking and feeling systems
without uh worrying about the first
person experience side at least in this
paper
and then the
key technical question of this paper
which
broaches onto
a million and a half other areas is what
is a markov blanket just what are these

blankets

and how is one modeled identified
defined statistically what are the pros
and cons of different approaches and so
that's some of the technical details
that we'll get to so steven thanks for
raising your hand and then welcome
serval after that feel free to say hello
and introduce yourself if you'd like
yeah i'm curious and i notice here um
that we start with thingness which i do
agree is

this big question of thingness um is
something that

karl fuston brings up quite a lot but i
think he's often

he's the he in that part i feel he's
often a bit of a lone voice in terms of
really pushing the thing this side of
things and i think it's really
interesting that second question then of
okay how does that speak to system
sentience

and you see now we move into systems
right

and okay thing this sort of
to me is it gives rise to systems and
gives rise to sentience however

i think it's thing this is still there
right things are still the game in some
ways are being maintained so that first
question

because when we then get into the third
question things can get lost

and it's all about system sentience and

i think there's something really

interesting there because

um

there is a question of i mean it's it's
even as we say in those brackets what it
really means to have system sentience
or even just system is
probably rather ill defined and and i
suppose to some extent is how much is it
even needed

how much can thingness be maintained
as an equal partner in the game and not
sort of passed over once we scale so i
think that's a good good piece of
provocation

thank you and to what extent can things
and systems be identified in a bottom-up
way from

measurements of dynamical systems
serval would you like to add anything or
say hello

or bonjour

hello i'm turbo

i'm a complexity scientist

i founded the chemistry social
laboratory which is interested in
building uh

inactive models of social political
changes and i'm here
to

see the discussion around the formalism
of the paper which i did not
revise and i therefore cannot make
informed comments on

thank you sir

dean

i just want to add one thing to what
stephen was talking about there

i think what

the big questions and what this paper
addresses pretty directly is

can we

can we accept a minimum of two that

there's an abstraction

and that there's a material

aspect to thingness

and one and marrying those two things

is is kind of what

we're hoping to be able to do but but it

it comes in two very discreet forms

so

so

what of that

we call it thingness because it kind of

spans

both realms

and that's what i think i take away from

the idea that it can be chaotic

and not

absolutely there's the map in the

territory

and

one thing that we're gonna return to is

like

the first map that we apply to the

territory so already we're separating

the territory in the map but even the

first map can be chaotic

and then can we put a second map

the laplacian approximation as it turns

out in the paper can that second map

be one that gives us some properties

that make it more tractable to handle

like

you know potholders taking something hot

out of the oven give a better grip

on the first map which gives a better

grip in terms of actionable insight into

the territory can that second level

both be more tractable especially using
computers but also retain those chaotic
properties which we're going to see when
they estimate the lyapunov exponent of
the laprosian approximation

so

that's the recursion prior to the rep
the repetitiveness

that's what i think this really gets
into recursion modeling

thanksgiving so stephen

yeah i'm gonna put this as maybe a
little bit of a question as well but

it's rhetorical if it doesn't want to be
answered but i think that

the thingness and the markov blanket
because we're getting to that stage here

um

there is this

there's i think this tends to be a sense
that okay we have a thing and because
it's

it's trying to sort of maintain itself
in a just

dissipative world

then

that's effectively a system

because it's got a markup blanket and
we're in that world but with this it's

like look you've got lorenz attractor

then lorenz attractor goes into

thingness

right

now

what and then thing this potentially
goes into

probably with a few phases system

sentient system so the question is which

side of the which side of thing this
is markov blanket
does markov blanket emerge
from him from going from lorenz
attractor into thingness
and then continue or does it is like
lorenzo tractor into thinness into
markovian
blanketed thickness
um
curious what your thoughts are on that
and maybe this paper shed some light on
that
great question let's look at the road
map
and just
see how that comes into play
so we talked in 32.1 about a few
different ideas and we have some slides
ready with just
tabula rasa if we'd like to add notes or
we can always modify slides and deal
with questions as people are raising
them
let's go to the roadmap though okay
so the the roadmap of the paper is as
follows
in section one there's an introduction
in section two
there is
from dynamics to densities so this is
one of the really fundamental kind of
linking events early in the paper
where dynamical systems
dynamics
are linked to flow systems like
densities
so

this equivalence between dynamical systems and density driven systems is going to be finessed extensively in the paper so that's an introductory step that's very important to note the helmholtz decomposition is a tool from vector calculus that applies very well to densities and flows where it's usually applied but because of what links were made in section 2 it becomes possible to take that helmholtz decomposition approach to dynamical systems that is specifically visited and revisited and then moved beyond in the lorenz system that's one of those classical chaotic systems and a favorite toy of complexity analysis the lorenzo system is moved beyond and then approximated with a laplacian and then even moved beyond there so that's pretty much where we got to last week we tried to do a really solid grounding in first the equivalence between dynamical systems and density driven systems or two kind of representations of underlying systems that could be seen as equivalent and then we talked about the lorenzo system and about how it's chaotic and how chaotic means like sensitivity to small changes as quantified by the lore by the lyapuniv exponent and then we talked about how the

laplacian approximation which is kind of
like fitting a quadratic which is kind
of like letting the ball roll to the
bottom of the bowl and this idea of like
well if we can phrase the problem like a
bowl

then all we have to do is roll to the
bottom there's only one bottom
and we know which slope we're on so
that's also called convex optimization
because no matter where you are if you
know that the problem is shaped like a
quadratic curve like a bowl
you can just instantaneously estimate
the slope of the bowl and then just run
downhill

whereas if you're on a rugged landscape
just you know you're lost in a mountain
region we'll go downhill
you're going to get trapped into a
valley potentially and you might not
make it out to the ocean
but if you could re-imagine that rugged
landscape in a way that was more bowl
like it would allow for
tractable

estimation of which way to move
in section 4 we get to the markov
blanket concept and the free energy
principle and i think that's going to be
the really fascinating discussion is
like

okay so let's
get what that markov
blanket what is that partitioning what
are we really talking about
and then are we talking about
the territory

are we talking about the lorenz map or
are we talking about the laplacian map
where is this blanket partitioning
coming into the picture
and then how does that help us reduce
our uncertainty about systems
and what does it say or not about that
underlying system and that's where we're
going to be talking about particular
partitions

of course a favorite and recurrent pun
which is that particular means specific
but also it means like a particle like a
autonomous particle drifting around in
space

and then section five
really comes to a
culmination

with a discussion of the free energy
principle alone
and so i think it'll be interesting to
discuss like what is subsumed into the
free energy principle and why do we need
one section mark on blankets and free
energy principle and then another
section on just the free energy
principle so that's kind of the road map
steven and then d

yeah just a quick question see if this
clarifies something which i think is
what you're saying

dynamics is um

the dynamics more particular flow like
it's and densities

while you could have a density of
particles it's actually more like a
probability density

on wave functions would i be right in

thinking is if there's a kind of a wave
particle
or part actually a particle then wave
way into this and that's
partly how they're
playing this out
someone with
more experience with the formalism would
be totally open to their
contributions on that question my first
take would be
you could make a dynamical systems model
like equations of motion for
a particle floating around and so it's
like you're focusing on the first second
and third derivatives of movements
in x y z space or tetrahedral space for
some particle so a dynamical system
model could be describing the movement
of a particle
but
the um
flow that gives rise to the movement of
that particle is it's kind of like
figure and ground like the dynamical
equation features the movement of that
particle like i want to know the
pendulum's location
and so that's going to be a dynamical
model of the pendulum whereas the flow
based models
like you pointed out are much more about
well what is the underlying space of the
flow of that pendulum so it's like if
it's to the left of the bottom it's like
it flows this way and if it's to the
right it's like it flows that way and so
that is about asking what is the

underlying flow dynamics that do give
rise to movement of potentially
particles
and those are why this meme is on the
slide and why they are made in
equivalence early in the paper because
it is like talking about the difference
between the movements of a point through
a space
versus
understanding the flow dynamics of that
space that yes does give rise to the
movement of particles
ah and could i just ask one so what you
almost it goes from a
just purely the um you know the momentum
of a particle to this flow
and the flow is kind of a bridge then
between density dynamics which is maybe
more of a probability density of
where it could be which can map both to
an object where an object might be but
also where a wave function might be
in in broad sense so then you're
sort of going from literal particles
in a classical sense to the flow
and then the flow then bridges into
density dynamics which is can be more
probability orientated is that
fair to say
yes
particle is going to take a particular
meaning a little bit later when we talk
about the partitioning of states but i
believe that's correct and also the
density
like you suggested
ties much more closely to statistical

density distributions like in statistics

we're often interested in density

distributions

of special type like the area under the

distribution summing to one the integral

over a distribution being one

allows us to treat it like a probability

distribution and so that's the whole

debate in quantum is it actually a wave

or does that

distribution

of positional uncertainty merely

describe our statistical inference on

the location of something that's at a

specific point

so

whichever side we want to come at this

from

section two says it's gonna be okay to

use both and we're gonna sometimes lean

more on our left leg and sometimes we're

gonna lean more on our right leg because

there's tools that exist for one but not

the other dean

yeah i don't know if this

begins to answer steven's rhetorical

question but i think

where

most people get introduced to a markov

blanket they get introduced to the idea

that it's either a way of partitioning

and dividing or it's a way of being able

to envelop or blanket and i think what

the paper points out is that it's both

all the time so you can for example find

a box of cookies

and see all the cookies settling at the

bottom

whereas or if the person is you know
described as being creative
uses that box of cookies as a leverage
point to get more cookies off the top
shelf now it's a platform
and now it's a mound
and so we can see it both ways at once
if we think of it as both ways at once
it's not an or a question it's that
minimum of two piece again that that is
really fascinating about this that
really kind of blows people up it's
really both at once then you decide
which it's going to be
and that's what i think is really
fascinating for people who are just sort
of start getting into this because they
they've been so habituated to saying oh
it's it's a cookie jar
oh it's a platform no it's both
and that's that's what we want to do
thanks dean and um marco in the chat
wrote like a boat in a stormy sea it
moves at the mercy of the seas and winds
flowing forces with only its rudder
sales and other limited mechanisms to
steward the world's influence on its
journey so bucky meme one unity is
plural at minimum two bucky meme two on
his gravestone call me trim tab kind of
like the rudder and that is exactly
getting at the point of this paper which
is if we just were talking about a dust
molecule floating in the air then we
could say well we could use the
dynamical systems model
of the position of the dust particle or
we could do inference on the flow of the

room
and that gives us a different and
complementary totally co-extant picture
but that's a dust molecule
what about active systems that have
affordances that are able to
do inference on action selection
choose different policies
how do those active systems
do inference that's the winged snowflake
that's the moth to the flame and those
are the really exciting systems and that
also speaks to the difference between
what's called mere active inference like
the particle
where we can still have this
partitioning and still see it as a
dynamics or a flow versus the adaptive
active inference which is then we're
thinking about those systems where the
particle the particular states
are able to effect different policies
and select between them
in a way that acts as if it's resisting
dissipation and those are the kind of
exciting systems that we want to get to
by the end of the paper stephen
and
pulling on that idea of um
cookie jar
is the there's the the cookie jar is
maybe a almost a particle a thing
that is like leveraged and there's a
thinness
which it becomes
so it say it has a thing about it but
there's a thinness of of being
an affordance

so this is an interesting just sort of
came to me but the thing this in itself
is slightly emergent because there's
the the only by the process of trying to
use it to get the higher cookie does
another kind of thing emerge which
becomes a step
so there's something interesting there
about the nests
um as as an emergent you know is it ever
a step
really
but it's always
it's never not a thing
at the same time so there's a i think
there's something quite interesting
there in that emergent stroke
flow stroke you know literal sense
and also
i don't know to what extent this is um
intentional but thinness
we talk a lot about ness the
non-equilibrium steady state or
non-equilibrium stationary states and so
it's like the things
ness
that's the thing the thing we're talking
about is the thing's ness
so we're not talking and saying that the
system
is at stationarity we're saying that the
things nests
by virtue of how we're modeling it
has stationary characteristics
and that is the thing as we model it
which is that
always existing difference between the
map and the territory we're not saying

that the territory is even a thing we're
saying the way we're relating and
interacting with the map which we
totally made no one's going to deny that
that maps
things-ness
is tractable i can hold the map in my
hands that gives me better grip on my
road trip
steven
that's a really
yeah i think also the
the thickness can tie into
the idea that something
ontologically
comes into being through relation
through some sort of network dynamic
um in which it acts
um
and that that actually that speaks to
this lorenz attractor you know so the
lorenz
becomes part of a network of an ontology
of thing which is really a thing in
relation to what some other thing is
like
you know because obviously the scale of
the cookie box is quite a lot different
to the scale of the atoms in the cookie
box or the
whatever scale we want to so there's
that and then there's some sort of
affordance level
and maybe we can get out of the whole
sentient system thing and it's like it's
just an affordance for the thinness to
maintain thingness
it doesn't need to know about systems as

such
unless someone some creature happens to
evolve a brain which starts to sit down
working out system diagrams
two-legged ones potentially
um
otherwise it's like it's just things all
the way up right
let's
now that we're
30 fun minutes in
let's um sprint up to that
markov cliff and then see where we go
so
figure one of the paper
is
kind of making this first fundamental
connection between the lorenz system as
defined by dynamical equations so
the traditional lorenz is defined more
like the blue version within equations
of motion
um well it has a positive lyopine
exponent which is to say that very
closely
located points diverge rather than stay
in like a laminar flow parallel line
world or converging so they diverge
and that's what makes lorenz system
chaotic ergo difficult to predict
measurement error and tiny perturbations
due to even thermal noise which is like
kind of how pseudorandom number
generators work
and then
there's going to be an approximation
based upon this stochastic solution
and so the trajectory in the right panel

here it's the deterministic solution to
the laplacian form of the lorenz
based upon the helmholtz decomposition
and so this is sort of like the version
on the right that's usually seen and the
state space model is almost like we're
sampling stochastically from points
along that race track

so that was

figure one

we could you just could you just explain
that just go over that one more time
just to clarify that bit i think i
understand what you're saying but
it's kind of an important point you make
there

when you say deterministic solution
that i didn't

there is that that went just to clarify
what that means

so the deterministic solution
would have no

noise

so that the

position at the following time is
entirely determined

by the

non-stochastic

state that it's in at the preceding
moment

whereas when we talk about a stochastic
form

it's like

that's like your car's driving

and there's thermal vibrations on the
road now

the leopard if exponent on that car at
non-zero temperature

is small so that the flow term dominates
the vibratory term but there could be
some other system where
the difference between well if it's in
the absolute zero world and just on the
road then it's going to go straight but
once you introduce any
wiggling
when the system is chaotic it has that
positive lyapunov exponent then the
tiniest wiggle causes the system to
diverge
really helpful because and so basically
normally we just like with the
deterministic thing it's just whatever
the initial conditions if it varies
slightly but then once it runs it will
settle into something but if you
you've got the initial condition and
you've got the perturbations due to
noise
as it's actually running which adds in
suppose you could say a little bit of an
extra
change in initial conditions to each
state that the Lorenz attractor is
moving through so okay that's that's
helpful thanks Daniel yes and I'm gonna
just show this image this is um
I don't know if it's the very very first
but this is the image that Lorenz
used to talk about chaos in 1961.
so I mean what's the difference between
76.8 and 76.85
it's imperceptible if your thermometer
only had like three significant figures
you wouldn't even be able to talk about
the difference but I mean how different

can it be

and it turns out that these those two
systems start imperceptibly different
but

nearby points in chaotic systems do
diverge and so it's like okay but
they're still tracking each other right
it's like well no wrong

later on and so of course a more chaotic
system with a more positively open of
exponent diverges faster but any system
with a positive leading leopard of
exponent is going to have divergence
and so

if you have a purely deterministic
system

or one without this kind of the openiv
induced chaos

then you can predict far out

however

prediction matter experts as dean i'm
sure will revisit us too

meet their match

with chaotic systems and that's actually

one thing i hope that we get to by the
end which is that by putting action in
every loop and inference on action

we can still whichever of these
timelines we end up on

we're going to be doing okay

because we're always re-adjusting

our inference to whichever of these

mountains we're on it's not like one of
these is better or worse or more or less

challenging like they have the exact

same upper and lower bound

the only mistake would be to make a

prediction and be super confident about

it and be on the wrong predictive
timeline

but as long as we're always updating our
predictions

then

either of these days are gonna be okay

but if we put on the jacket because we
thought that it was gonna be cold here

and that's when it was hottest then

we're injured

but if we just think okay well we know

we have the jacket afford in so we're

just gonna keep an eye on that as we

watch the temperature and continue to

update our predictions maybe then it'll

be okay

so that was the

initial lorenz

attractor

section three goes into the helmholtz

decomposition which we've talked about

in several other streams

and um

we've seen it introduced as the

breaking down of a vector field the

total current in this kind of

electromagnetic representation

into the irrotational

which is kind of like the straight line

there's no rotation in those straight

lines and the purely rotational

solenoidal current flow currency

we've talked about solenoidal ingredient

partitioning and then what this paper

does and i think

the technical details are in the

appendix b

of 16. so that was the bayesian

mechanics paper
and then also there's this housekeeping
big lambda
and that's in one of the appendices to
this paper so that would be like
something for
someone else to help us understand a
little bit of because it does matter
and it's actually a key contribution
because
if
the landscape is changing moment to
moment
you can't just do the helmholtz
decomposition once
and sail on that c
without updating
your estimate of the partitioning here
of the decomposition here
that
decomposition so again dynamical systems
that's like the position of the car and
then we're going to because of section
two be
having an equivalence between that
dynamics and the flow modeling
and one of the terms that shows up on
both of these
is the fancy j fancy i it's the self
information
the first part performs a riemannian
gradient descent
on the negative log of the steady state
density which can be interpreted as
self-information so how surprised should
one be
to be at a given location relative to
expectation so we talked about how like

the potential energy of the ball higher
up on the bowl is higher
and then the potential function drops
and that's the physical analogy to this
more informational statistical
distribution approach to
the self-information
and then in 3.1
that
self-information is going to be pursued
and expanded upon for
a a relatively more simple case
so we're going to identify due to the
helmholtz decomposition that there's
this key variable which can be
interpreted as like a potential function
corresponding to the self-supprisal
and
that self-supprisal
term
may have functional forms
so we can have a function
that describes how surprised we should
be
stephen
yeah if you could go to that last slide
the previous one there that um
i like this is quite useful so
the the a plank if it's an
equilibrium
which we're not at because we're doing
non-equilibrium steady state but if it
were then it would be zero basically
it's that equilibrium so it's not
changing so
there's no there's no divergence because
um it's zero
and then he's taking this helmet

composition which in a way is like a
vector isn't it with a vector you've got
you know you can split something's
moving into x and y so the this the
gradient is coming out and the solenoid
was at right angles to the gradient
so it's kind of like that trick that he
does with
accuracy and complexity
he can
say that there's a solenoidal and
gradient which go together
for when it's still in a kind of a
particular
angular momentum sense
and then there's that ability to
with the housekeeping
to then move it into the flow
which i'm not entirely sure i think
that's the bit where we might need some
quite what because flow could probably
be more specifically thought about
so now you've got this ability to
sort of sort of structure
through the kind of more particular
context the solenoidal and gradient
and then there's this transition
to a more kind of general
variable
which can be just we don't need to get
the particulars now because it's a
variational
metric that just gets used in an
approximation sense
and
then it follows into what you're saying
would i be
on the right tracks with that

um

not sure who he is even though i have an expectation because it's a multi-person paper but just looking at these as like circles and triangles and squares we can see that upside down triangle fancy i of x is in both terms so there's like a q multiplied by something and then a γ multiplied by something and we see that ω is defined as like the q minus the γ so these two terms which were decomposed out from each other get recombined into the final flow

so

that's just like if it were like a times x minus b times x and then ω we're defined as a minus b times x

so we're

recomposing

a flow equation that still features this

fancy i

and yes we're solving that for

a specific

point that matters we want to solve that

for where the ball is at the bottom of

the bowl

so there's

a function

that's going to tell us more or less

in multiple ways

whether we're

more or less

away from the bottom of the bowl

and again it's not about the system

being at stationarity

it's about this decomposition

having a focker plank steady state

dean

just one quick thing so at the bottom of

the bowl if we the gradient can

effectively be zero but solenoidal could

still have a value because it can still

be spinning around the bottom of the

bowl but the gradient then because it i

noticed the gradient

term

isn't in the flow but the solenoidal

term is

the gradient is because q minus γ

is in ω so ω is this ω

del fancy i

is just these two terms combined okay

okay yeah dean

yeah i was just going to say what

typically when this is brought

forward it's easier to understand the

decomposition and the recompiling if we

view the blanket as a partition where it

gets a little bit more interesting is

when the blanket is literally an

envelopment

and there's a there's a

physicist by the name of keith humphries

he talks about emergence and he looks at

that piece of it as well because there's

a polar aspect to this we can think of

that in terms of positive and negative

uh fields magnetization and stuff like

that and

how that is also a decomposition and a

recompiling

so i understand how it was presented in

this paper

that's the partition way of looking at

it right it's more or less but there's
also the
positive and negative polar piece to
this when we look at the markov blanket
as a blanket
as opposed to a separator and again if
we're holding up two things at once
this this doesn't
this doesn't change that it's just
making sure that we hold both things up
at once the envelopment
and the divider
i just want to bring that out there
because again
we can we can see it easier as a
partition
like the skin as a partition we don't
typically see the skin as a platform for
sunscreen
that's my point
thanks dean stephen
yeah
this this this um this is a good point
in terms of
you can imagine
at first i was thinking about grading
you know it's obvious that you've got a
gradient that you can pass through
maybe where the gradient's zero but also
solenoidal flow i suppose if if things
are moving around this way and it starts
to turn and it starts to go the other
way at some point
the the solenoidal flow is also passing
through
um a zero so they can both pass through
a zero point it just wasn't as intuitive
when i first thought about it

yeah and even if you have something
that's spinning around like something
that's spinning in a circle like a
hammer thrower it's actually continual
acceleration that doesn't mean it's
speeding up it means that the vector is
continually changing

so now our map

is of that acceleration and it is
unchanging in how it changes
so that

is to say that the system can have
at a

in a different way of thinking about it
yes a solenoidal component we're not
saying that balls at the bottom of bulls
are dead and unmoving

we're saying in our map

there's only one point because we set it
up to be like a bull map

where the derivative is zero that's the
solution where

the rate of change is zero

so that's not to say that the system
isn't moving that's to say that the flow
steady state has been achieved
in figure four

well in figure three

we see

um the three states so this is like the
cover of the geb book

like kind of three different projections
from a three-dimensional system

um the second versus the third the first
versus the third and the first versus
the second

and we're seeing how there's
movements

of sampled trajectories
and they're kind of coming back to
something that's bowl like for any set
of two dimensions
so it's kind of like a three-dimensional
bowl now
because any two of these dimensions
it's more like a regular
bowl but the kind of soup goes in but
then
like the the soup bowl is like
a down projection of a system that has
an attractor in a higher dimension
potentially a much higher dimension but
lorenz they're only modeling three
because the particle only moves in three
dimensions
when we continue to investigate
the bowl-like properties
of our laplacian approximation to that
lorenz system
we see that there's a few different
shapes of bowls depending on which angle
we're going to shine the flashlight from
some of them are like spherical
so a spherical bowl doesn't have any
specific correlation
between
the two dimensions
it's like a cloud of points the
regression is just like flat through it
however there are some
couplings
where the bowl is non-random so it's
almost like if we were tracking the
movement of a ball it's getting jostled
around in these bowls well like the one
of the first and third if that was the

light we were shining it would just look
like it was a scatter plot this is
totally the regression's flat there's
not much information being provided it's
it's
within the bowl so it already has some
nice dynamics but it's kind of within
that attractor zone
it's a scatter there's no pattern
but clearly for the first and the second
we see that there is a coupling now
imagine if we didn't see the ball
because of course we don't it's not
there that's part of our map we're just
seeing the movement of that ball as we
sample it through time
and it would be only on this manifold
kind of being like spending more time
going from the top left to the bottom
right in these images
so we're going to take that idea
and then connect it to the jacobian and
the hessian
which is again
throwback to our matrix math slide and
um an awesome opportunity for anyone who
wants to bring more of a technical
interpretation to the stream
the jacobian
is like
the
first
the the unnamed matrix is just like the
matrix of position
so that's like
the first moment of the distribution
and then higher moments are like the
partial derivatives so jacobian is like

the first order partial derivative so
that's like the slope of the ruler at
that point and then the hessian is the
second partial derivative that's like
the curvature so if it's a bowl there's
only one point where it's flat
and then it could be flat and the bowl
could be like up or the bowl could be
upside down and so that's why it's
important to also have this second
curvature term to understand whether
we're at a bowl like this or a bowl like
that so that's kind of like calculus
and
those
bowls those bowl attractor shapes which
we're inferring using the laplacian
quadratic assumption so it's not saying
that the lorenz has these possesses
these is these it's saying that's the
approximation that we're going to make
it turns out that although there is
somewhat of an unclear
correlation pattern in the first
derivatives
so like the movement to the left it's
not like it's correlated with movement
up
however
the second
partial derivative the hessian matrix
starts to look
a lot like the covariance matrix
so the actual positional covariance
of these states with each other so like
the on diagonal is kind of obvious like
of course your position on the x-axis is
correlated well with your position on

the x-axis
but
there are
correlations between state one and two
and
that is recapitulated in the structure
of the second order polynomial
approximation to these attractors that
are laplacian approximations
so
again up until that point in 3.5
and in figure 6
markov has not been introduced
nothing has been said
that word hasn't come into play it
didn't come into the system's definition
and so that's just really important
because it's just showing like
there's a lot of groundwork to get to
where we make the markovian partitioning
and so
there's
power and challenge in that stephen
would i be correct in saying that
effectively it's a it's a case of
approximating
the
states
of the lorenz attractor
and that's what's that's currently there
so this is different ways of
showing that within some manifold state
space
and
by approaching it in different ways it
reveals a different type of
approximation the only thing i would ask
i'm a little bit confused why it does

squish maybe i'm missing something i can
see why
there'd be kind of a round one
i'm just trying to work out how it
squishes is it something to do because
of the solenoidal piece or the gradient
piece or is that something else
okay
so
if we were talking about
just a particle undergoing brownian
diffusion in three dimensions
so
it doesn't have any kind of correlations
or any sort of structure just it's just
a random walk in three dimensions well
then it would have a spherical error
profile for all of these images
but
we are capturing structure that exists
because lorenz attractor has a specific
specification
so the states are whichever are being
modeled if you cared about pressure and
volume that's your state space that
two-dimensional state space of the map
it's not saying those are the only
states of that system it's just those
are the ones that you're modeling
in the lorenz system
we have three states
it's positioned in three dimensions
but those three dimensions again also
apply to a brownian diffusing particle
which has a different
jacobian covariance and hessian
we're talking about those being
calculated for

the lorenz system with the parameters
that they selected
and so we are pulling out
covariances
and connecting it to this second order
polynomial approximation in the hessian
for a system
where the movement
is not just a brownian diffusion and so
that's why there's structure in the
covariance and that's why the structure
is interestingly recapitulated in the
hessian dean
and all i would
all i would supplement that with is
things things
there we go things that are
in
something
that's moving
the thing that's moving that's in
something that's moving that's on
something that's moving has a tendency
to have
the
negative charge things all congregate in
one place down in the lower right corner
and all the positive things all
congregate in another place in the upper
left corner that's the nature of
movement
and so
that polar part of it is in play as well
if we
consider it as part of these statistical
transitions and that's
again again
it's it

it's fascinating if you if you look at
it a minimum of two things so not just
not just as a partition but also as a an
an envelope those things suddenly
pop out at you and you see the pattern
in in a different way
so i want to think about a physical
system that's not chaotic but will help
us understand what this covariance
matrix is so let's think about a
children's carousel
so it moves the horsey is moving around
in a circle and so that's like our unit
circle like the four quadrants and then
also the horsey is going up and down
okay so when you're
you know so again it's like you know if
you're at 12 o'clock on the carousel
it's like you have a one for the y and a
zero for x okay so i hope that makes
sense now as it goes around it's like .5
and 0.5 again just using simple numbers
here like the x and the y are correlated
and then they're also correlated here so
one could say that there's like a
correlation between the x and the y like
they have a covariance together
but now the z-axis is moving up and down
independently so a system like that
you'd see
a covariance between the x and the y
positions of the horsey
and
not that they're a different system
but that they are
uncorrelated they don't provide mutual
information on the z-axis which is
oscillating now

if this paper had done we model a
deterministic carousel
and it's kind of like putting the rabbit
in the hat because the covariance
matrix it looking like this
would be no surprise because it would
have been defined that way so that's the
key insight and one of the key
contributions is like
they didn't put this in the hat
this is
the og complex system
yet
the approximation
the laplacian approximation
pulls out a covariance structure
that's extremely tractable
and
it turns out that it still has a very
strongly positive and very similar
leopard if exponent
so it's like now it's like a chaotic
carousel
and yet our approximation our map of
that chaotic carousel
still holds the chaotic nature of the
system
while also giving these bowl like
attractors
for any given two and even all
dimensions or states considered
simultaneously
so
um
we're not conflating the simplicity of a
non-equilibrium steady state with the
complexity of the underlying density
dynamics

so just because the map is simple
doesn't mean that the territory is and
so that to me is a knockout blow
against realism and for instrumentalism
because this paper's saying nothing
about the underlying system
active doesn't say anything about the
underlying system we're talking about
approximations and our maps of
territories

in other words

the probability density can evolve in a
complicated fashion

in attractors

go ahead stephen

and

from what you're saying it's like if i
was to take that lorenzo tractor which
isn't

it's like a saddle often people often
talk about a saddle and you took the
the most um

agreeable

way of viewing it so you view it in in a
box space this way and look for how it's
moving that way and then flip the box
like this and see how it's moving that
way and flip the box like this and see
if it's working that way

those approximations would be those
three different distributions however if
you'd moved it by

like one degree

that that would be like

messy right so it's like they

if you look it's almost like this is how
the lorenz

attractor

taking the most um optimal
view of xyz
can be mapped out as it transgresses
through the saddle
um if that makes sense to bring it into
a physical space
um but again it's more complex than that
because this is taking it where there's
the plank is zero
to try and get
the most optimum sort of that's why
those things are
follow a nice
stable
transition whereas if you're taking a
sort of
an odd angle
it might all be completely a mess
would that be a fair
way of saying it
if you were to take a perspective a
projection down
that was a mixture of other
axes then
not sure what is gained by that but i i
see what you're saying there's some
particle moving in three dimensions and
you can have a projection into any side
of that cube or you could take some sort
of off kilter view that would be still
some type of linear
summation of all three so it hasn't
really reduced the challenge of
estimating but yes i mean that's one way
to see it dean
so i'm in complete agreement it has to
be instrumental
so that you can compare to the real so

the minimum of two you have to have
multiple maths not one
in order to be able to see this
and that basically then reinforces the
idea that the markov blanket has to both
at the same time be a partition
and be an envelope it's both at the same
time a minimum of two again
so again all we keep reinforcing is this
is this idea that if we over reduce if
we get if we get to a place of under two
we may be deceiving ourselves
and i think we'll come back to that soon
so now

one hour in
we're able to get to the markov blankets
okay

so

previously

we were talking about the case of a
single markov blanket and so if we
thought about like a bayesian graph so
now we're thinking about
three nodes each corresponding to the x
y and z axis

nodes a and b you know the first two
axes would have like an edge between
them because they have some type of
statistical relationship
and then it would be disconnected from
the third node because it doesn't have a
statistical relationship with them

okay

so now

to actually

start to get towards more like the
communicating systems or
system in the niche

that we're interested in that's the
target that we're trying to aim towards
in section four
the authors repeat the analysis of the
previous section
but approximate two lorenz systems that
are coupled to each other through their
respective first states
okay so we're going to look at this
matrix
okay so notice that there are two
this
make that transparent
so here's one
lorenz system
here's the other lorenz system
the the red one and the blue one
and then
there in these other
cells
these are the coupling terms
so
these are
couples this is saying that the first
state of each
are coupled so it's a bi-directional
coupling
and that it's saying like state one the
fourth state on the matrix like matrix
row and column four
is the first state of the blue system so
the first state of the blue one the top
left of the blue and the top left of the
red
are coupled through the off diagonal
green
boxed terms
if this was all zeros nine zeros here

and nine zeros here
we would have
two
independently evolving lorenz systems so
we kind of took the lorenz system
the three by three matrix that specified
the dynamics of that system
and then we just made another
now a nine a six by six matrix
with two lorenz systems that would be
evolving
statistically independently
unless we introduce this targeted
coupling
and they say this form of coupling was
chosen to be as simple and symmetric as
possible
okay
so this is like the next
step
but it's a difference that makes a
difference
because this is going to change the
total dynamics of it if you had just the
red and just the blue
you could like split that matrix
and you'd be getting the exact same
model ability because the systems
wouldn't be interacting
but now we've introduced these terms
even if we only had one
where you can no
longer divide this matrix into sub
matrices that perfectly describe the
time evolution of the system
so 7 figure 7
is like figure 3
so it's similar we're taking like sort

of pinpoint
locations on
the laplacian approximation of the
lorenz system
and now seven is doing the same
but it's looking at these two
synchronized systems or maybe it's
better to say that
there's two
coupled systems
entangled do yes and due to the
structure of their coupling
it results in an entangled movement
pattern
such that their trajectories have very
um
[Music]
have their trajectories are informative
of each other
and that's also reflected by this flow
diagram where clearly
their
movements
exist on a manifold
that's in a restricted space of the
possible so two people brownie and
diffusing in two separate rooms you'd
get that circle
scatter plot
okay two people who are tied to each
other
you would see a perfect correlation and
so we're seeing something even though
each system has complex and even chaotic
endogenous dynamics we're seeing that
the introduction
of this coupling term
allows the systems to actually

empirically

have

entangled behavior steven

we've now got lorenz systems as opposed

to an attractor

and i'm curious when does it go from an

attractor

to a system

is it because is that does the coupling

effectively start to create

more systemic

nature or you know i'm just curious

about that

is it because it's the

systems the lorenz system is just these

three this is the specification of a

lorenz system and it has an it's a

chaotic attractor and so now we're

talking about two of them

so there's two coupled lorenz systems

but it doesn't make sense to talk about

two coupled attractors because that's

not what's coupled the systems are

coupled as per the matrix specification

and then it doesn't even make sense to

couple the attractors the whole point is

there is an attractor now

by virtue of the coupling of the lorenz

systems

is it almost like once you start to have

something and you're going to relate

them

as soon as things relate there's a

system

because there's a relational quality to

it

whereas previously it could be a

mathematical for

formalism it was still described as
lorenz system so i wouldn't get too
hung up on that word but yes systems
have multiple interacting parts
okay
figure eight just like seven
revisited the format of three
eight is going to revisit four
and
now we're going to get to where we start
to see the partitioning
figure eight's gonna revisit four so
remember the first partial derivative
the second partial derivative in the
jacobian and the hessian respectively
and then the covariance so we're gonna
do that same thing that we did on a
three by three
but now we're gonna do it on a six by
six
the log jacobian looks a lot
like
this
the non-zero cells in the specification
of the system
which
is the system itself for systems that we
get to specify
it looks a lot like this but again for
territories we don't get to specify
the map looking like this
does not say something specific about
the territory
but it's not super surprising that um we
actually see
like similar this top right has
like no correlation like one and three
are not correlated and we see that

little chip missing here and here
so we see
a lot of the same dynamics at play
within each of the coupled lorenz
systems
but also we've introduced
a coupling
now that coupling
induces a new and different covariant
structure
look at these gray squares in the off
diagonals
now one and two of system a
are coupled to one and two of system b
less so a little bit grayer
than the one
that are within each system
but still very marked bi-directional
synchrony is being induced
the interesting piece comes into play
when we look at this second order
approximation
and that's where we see that several
cells can be identified so here we have
like the magenta mute please stephen
thank you the magenta and the red
are those first states of system a and b
those are the ones that have the
coupling on the fourth okay
then you have the other two states of
system a in teal and the other two
states of system b
in dark blue so it's almost like we have
four kinds of
states
four kinds of states out of these six of
course you could say well each each
state is unique

great totally true they're different states that's why we have them as different dimensions in our model but there's going to be some interesting things that fall out as a function of us separating out the two systems and then separating out parts of the system that interface internally and externally and then parts of the system that do not directly by virtue of how we know we specified the system do not couple out again the fascinating thing is that like state two of system a and state two of system b have a covariance that's like maybe a mirror neuron again using a more like realistic mapping it's like there's some internal part that is only interacting with other internal parts by virtue of how the system is defined and yet these two parts in each system which are not directly interacting by virtue of how we set up a system have covariance so that is what leads us to this numerical analysis which kind of provides a little bit stronger and different empirical evidence so it's not

just an analytical
formulation it's um
a numerical simulation
where
we can see that when we actually let
that system play out
that even when
states
start very correlated
like we just start the third and the
sixth state in a very similar position
so they start out with a high partial
correlation coefficient
when we run out that simulation through
time
it ends up trending towards zero partial
correlation
so we're recapitulating
the correlation patterns
of
the system
even when it starts
in a more
correlated place
and so that suggests that some very
essential
aspects of the system
are being isolated and tractably handled
by this kind of partitioning of coupled
systems
so it's pretty interesting
now we've seen this top oh dean yes and
then we'll just grow quickly danny what
it also says is that hype
hyperbolic
is also existing in this pattern
although it's not necessarily being
materialized in these diagrams it's it's

also there i mean what that's what this
explains is that there can be continuity
there can be alignment and there can be
the hyperbolic
function within

all at the same time again that's i just
wanna again i just want to point to the
non-obvious but it's still there

great

so we've seen this top of steven yes
yeah it's also going to add that so by
showing the nature of the coupling like
varying the nature of the coupling that
you showed

it's sort of reached it can reach a
different sort of
state

non-equilibrium steady state in some
ways would that be true
there's a different pattern emerges as
so

it could speak

maybe i'm over extending this but
changing the nature of how things couple
could in a way change the sort of states
which things can fall into so use you
could almost have multiple states
being inferred based on how the coupling
is

set up oh yes i mean we would throw out
any model that treated two people tied
to each other as simply moving the same
way as two people in different rooms it
is the coupling of systems

that

influence their behavior

and again we're not talking about the
causal structure of the world territory

we're talking about the models that we
got to specify
and how changing the coupling pattern
changes their behavior
dean
yeah and stephen when you're taking the
abstraction here off the page
so you could set up your programming for
example that says okay
i'm willing to sponsor somebody
because i want to mentor them in
bricklaying you can start with that
or you can start with
there's two people that are interested
in brickling but the first thing we have
to address is whether there's a
potential for positive entanglement
that in real terms that's how you can
flip the
the abstraction
into the
the likelihood
of those
entanglements turning out
extending because the people will
actually be relatable to one another
so that i'm just again i'll bring that
up because
that's how we turn the abstraction into
the
so what does that look like
okay stephen something quick on this
otherwise we'll get to 10. oh yeah just
very quickly yeah so that that speaks to
because we often think about the
the
the model on the inside the generative
model and the external states and

you know the model trying to understand
what the external states are doing but
this speaks to
the ability to
to change the way the blanket
is coupling so that you know how much
work is going on
in terms of
enabling there to be information
variational free energy information
being used in the generative models how
much is being done by actually changing
coupling
dynamics in the blanket
is that make is even feasible that you
know because that in itself will start
to yield more and more ways for the
generative model to start to extract and
make inferences based on
what's going on externally so that that
also helps i think so as a heuristic fat
pump yes great but actually the blanket
is downstream of the coupling pattern so
it's like changing the coupling of a
blanket is a little bit of a cart before
the horse because there is the coupling
of the system and that is what induces
partitionings
which we can then map to markovian
assumptions so it's like you know
when you put on headphones and you're
listening to this
yes a different blanket is in play
because the partitioning is different
because the coupling is different
between
other states in the world
dean on this and then let's go to 10.

yeah i'll be real quick so
like purses at charles sanders versus
abduction
doesn't guarantee that when you walk in
later and see beans in one hand and a
bag of beans in the other that you'll
get to
okay i just lost dean so i'm going to
continue
so
in figure 10
but he was talking about abduction and
c.s pierce
kind of that flipping that i understand
you're right it is the card ahead of the
horse
and sometimes that helps
not saying it's you and i know you're
you're agreeing with me but all i'm
saying is we shouldn't just assume
that it only goes one way
horse first card second
yes and in fact partitioning differ
imagining
differences ways it could be different
absolutely this is how we
are going to apply and ask what is if
for
active inference like what is
is like what is
the blanket states given a coupling
pattern well there's an answer
sensory states are the ones with
incoming statistical dependencies and
active states are the ones with outgoing
statistical dependencies those are the
blanket states what if
the coupling were different

then
what would the blanket be
so yeah it's not that that first blanket
that you identified is being changed
what's being changed is the coupling
which induces a different kind of
covariance amongst the states but it can
be almost thought of
in a cart before the horse way so just
to recap this nomenclature which we've
seen before
so we have external states sensory
states active states and internal states
 b is the set of blanket states so this
is one of the innovations of Fristen at
all beyond Pearl 1988 beyond Markov at
all earlier earlier earlier is that
blanket states
which in the
non-fep active world are the set of
states upon which
conditionally internal and external
states are independent
those blanket states again we're going
to partition them into incoming and
outgoing statistical dependencies sense
and action
now we have autonomous states α
which is the set of
active states and internal states
autonomous states are the ones that for
systems that we design we get to control
we get to control our internal states we
get to control our actions
we don't get to control directly the
sensory input
you can close your eyes
so that it's darker

or you can move your head so it looks differently and maybe you'll even be right about what sensory information comes in then but you can't just say i want different sensory information coming in so those are the autonomous states and then there's the particular states the particular states are the whole particle that's the blanket states which are sense in action and internal states so the particular states are like the whole particle floating around in the room or the whole cell or whatever again to give physical interpretations and then the autonomous ones are like the things that the cell can control or the thing that the system designer can control so we have autonomous states which composed of internal and action states blanket states action and sense particular all three on the right side and these are all being defined as a partitioning a cleaving apart from external states as modeled not of the territory of the map only stephen and as we uh mentioned here this this this paper gives a way for this to emerge from the very most basic principles and then we have like you described the the blanket states your autonomous states but of course in um any sort of system organism non-equilibrium steady state they're

always a nested so it's not like
the the the
there will be
a whole nested levels of markovian
blankets
so these when we emerge
at higher levels
we might be emerging
it's not like
unless we made our blanket there'd be no
blanket
there's going to be blankets all the way
up and down
so as a blanket starts to be formed or
brought into play
then
the
the nature of how that um those blankets
come into play
um
starts to draw in the idea of coupling
you know so maybe there's there's some
of those things that you just mentioned
there about
um
there's not a dependency the fact that
we're creating something from first
principles doesn't mean in the real
world
without the first principles we can't
make a blanket
because in in other cases it's going to
be sort of emerging through more
coupling between blankets because i
noticed we're talking about the blanket
but it's also good about the blanket in
this scale
and then the blankets in plural

at the multi-scale
okay yeah so within
internal states
again these are statistical dependencies
so this one can have other things
happening it's like a black box within
each one
and so
that is um for people who expect
to see the mechanics of the system
that's a disappointment
but for people who want to model real
systems including machines
it's a huge advantage because it allows
us to go in
keeping the map territory distinction
totally apparent
which is the real transparency we need
from our machine learning models you
know there's not going to be a
transparent explanation for how 500
variables interact
but we can transparently see potentially
how the map and the territory are
distinguished
now that's where we get to their
definition of markov blankets
so
let others
read through and work through the math
and critique it and build on it
but
we're talking about boundaries
based upon
sub matrices
of the total state space of the system
so like this is just two coupled systems
that we're talking about here where we

were studying like system a and system b
each is lorenz we we added the coupling
to one and four
and that's where we saw
a new pattern arising
why does a particular partition comprise
four sets of states
is it because tetrahedra are the minimum
polyhedra in our world
in other words why does a particular
partition consider two markov boundaries
the reason is that the particular
partition
again particular states blanket and
internal
the particular partition is the minimal
partition that allows for directed
coupling with blanket states
sensory states can influence internal
states incoming dependencies
and active states can influence external
states
outgoing dependencies
without destroying the conditional
independencies of the particular
partition
as shown in the upper figure upper panel
of figure 10.
so
very cool
how we went from one coupled system or
from one system with chaotic dynamics
made an approximation
and then showed that it had certain
well-behaved aspects and then we took
like the next step up which is to couple
systems in a way that we designed
and that

gave a new
manifold
that the systems were
acting entangled as if within
and then
here's where we get that very fun claim
that there's no claim
either the original lorenz system or
coupled lorenz system possesses a markov
blanket
we're not describing
the territory
i don't even think we're even describing
the first level of the map
the claim here is that there exists a
laplacian approximation to these kinds
of systems that in virtue of the zero
elements of the hessian feature markov
blankets
so this is one reason why you know while
also acknowledging that there's many
perspectives and so much to learn
that we do want to be
precise with our language
because it's easy enough to say that the
system has a markov blanket or there is
a markov blanket in the system or all
these other ways which prepositions and
nouns might be combined
and those might be so misleading it
could be truly incredible
so
it's a
word of warning
because
as we figure out exactly what the
technical details are
and

continue to clarify here it will matter
which natural language words we use to
describe it and it really does matter
dean
so
let me
add to that daniel so on an abstracted
level
uh we can we can frame the problem as
as in or out
and then when we translate that into the
physical space
we know when something
is entangled
or when something is different
but i think what this allows us for on
the abstract level is to examine
to sample
the markov blanket
as
in and out it gives us the potential for
a second way of looking at
how those two things work together the
markov blanket has both a divider
and an ensembler
do you agree
so so on an abstracted level it's very
important to be precise
but if we include this
we have to accept the fact that on an
abstracted level in and out
must be part of our
our repertoire not just in or out
because one is more precise
what is active inference while the other
allows for what if active inference the
and would you agree
i

think i agree

i'm just imagining that it could be

possible to design a system where there

was one coupling

going out but there was no coupling back

in

so

one could devise edge case matrices

that

have various kinds of properties

would those

even be good maps of systems that we

care about

probably not

so

for some of the basic attributes like of

communication of course if one person's

not wearing their headphones like yeah

the systems are coupled one person's

talking and the other person can hear

them but not return any information

right so it's like that's

interesting it's a type of coupling that

systems can have

and it has a different blanket because

of that

but especially for systems where we want

to think about states that are

influencing the environment and vice

versa whether it's communication or

stigmargy or coupling or collective

behavior

that whole class of systems

it's extremely important to have of

course in and out and more and less and

different

i wouldn't drop i wouldn't drop or sorry

steven i just got to get this in i

wouldn't drop or i'm i'm actually say
suggesting that we
we
this tells us we
must also
contemplate and so that's like saying
okay so which is more important that the
positive ions are the or the negative
ions they're going to congregate
and we can tell the difference but we
shouldn't apply a value all the time
there are times when we can just see the
difference and accept that for what it
for what it is as opposed to
always jumping to the idea that
something has to be
more or less that the positive ions are
more important than the negative ones
maybe they are but we shouldn't jump to
that right away done i'm done
okay closing thought on this steven and
then we're going to get to free energy
principle
yeah um
so yeah just just mention a little bit
of what dean said there like the the
yeah if something's a charged particle
how much is the system of the
molecule
showing the distribution of charge and
how much is it the quality of the
thickness of that thing right so i
suppose and and that that
becomes i don't think there's a
definite transition and and i'm glad you
went back to this diagram this is the
one i just want to mention there then
just to

that the arrows there on the right so
where century states
um it shows that the sensory states are
related
to the same external states as the
action states are affecting so it shows
it's almost like there's a there's a
regime of attention
um so it's not
the century states aren't reading
something coming in
about something which is completely
unrelated to you know it's not like i'm
looking i'm acting in front of me but
actually my eyes in the back of my head
right so my sensory there's there's a
there's an alignment
and and then
when i look to the the left
this this coupling that's going on then
between
in in the way that it's set up between
this
it's saying it's it's it's um
it's not a markov blanket when it's
saying it's a coupled system
so it's sort of saying in a way that
this more slightly more nascent version
there's this kind of um there's still a
spookiness about it i suppose we might
say it's is that this sort of a
it's not it's not it doesn't drop out as
far as a markov blanket as there's that
kind of
resonance going on and i suppose
somewhere between left and right
you start to become
you you meet the

the conditions of a mark of blanket and

i suppose

you know that's an interesting point

i don't want to just

make the water murky with too many

terminology we haven't brought attention

in it's not related to what we are

discussing here but i i see what you

mean

that yes

if actions were influencing state a you

know your actions were moving something

up or down but you were observing the

left right yes that's a different kind

of system than if you're observing and

acting on the up down movement of

something

that's not a regime of attention but i

know what you mean

let's get to the fep

okay

finally

remember that

edges

reflect statistical relationships which

is the other side of the coin of saying

that the absence of an edge reflects a

conditional independence

so

one can provide a definition of the

conditional density over external states

so probability density that's why it was

so important to go from dynamics to

densities in section two

and we're going to think about that

external density probability density

conditional statistical density as being

parameterized by the conditional

expectations of internal states given
external states
here's the formalism
see how
the
coupled systems have a flow
where like
given where you were on one even if it
were a little noisier as long as it
looked like an oval as long as you were
like you'd know that if you were low on
the left you were going to be low on the
up and down if you were further to the
right you'd be higher up on the y-axis
okay but imagine if the flow was like in
a circle you'd say okay we're at x on
the zero well
you really don't know because you you
could be low you could be high
or imagine if it was just like a scatter
cloud no matter where you were you would
like it was everywhere in the box you
wouldn't really know
but because
there's a manifold
then
this admits the possibility
of a
which means like sort of like
function-like and stretchable
map between the sufficient statistics of
the respective densities
not the territory
not the first map i don't even know if
we're at the second map anymore we're
talking about sufficient statistics
of densities
of approximations

okay

i think it's a death blow for realism

personally but i would love to hear

somebody who thinks it's not

the existence of this mapping

rests upon a continuously differentiable

and invertible map which is linear under

laplace approximation

there's a few technical pieces there but

it's almost like we set up

the system

in a way that was defined only by its

coupling

but then we also got to choose our

approximation approach

and it turns out that with a system with

interesting coupling dynamics enough to

befuddle almost every other kind of

model out there

the approximation

has a really nice relationship

that recapitulates aspects of the system

and

lets us map like if we're on low on here

we can go to low on here and low and

here to low on here like it's a kind of

it's a function that can go both ways

and so

that is

going to be

the relationship sigma mapping between

the internal and the external states

which was explored more

in the bayesian mechanics paper number

26.

this means the autonomous flow

can be expressed as a gradient flow on a

free energy functional of the

variational density

here are those four states four kinds of

states external sense action internal

and those are going to be expressed as a

tuple it has to be because it has to be

all four of those states co-evolving

and those have a gradient flow γ and

then the upside down triangle

plus

the solenoidal flow so remember way back

when

when we saw that we had q solenoidal g

γ for gradients

and then

λ for housekeeping okay so now that

is coming back into play

the solenoidal

is still a single term that's just the

potent the potential function that

self-supprisal

and

one

solenoidal term

then we have a gradient over

these four states and then the λ

housekeeping

that

can be rewritten

and now π

again

let

let the ontology develop so that we can

resolve some of this but notice how

particular states π

are the blanket and the internal okay

but

how have we usually used π

policy

action selection so this is a different
pie
the free energy functional on
particular states
particular states let's call them
equals
the self-surprisal on the particular
states
plus
a divergence term
that is the form
that then
lends
the ability to interpret that
free energy functional on particular
states
to be rewritten in several ways
including
energy minus entropy that's a very
classical chemical way to talk about
free energy
to be phrased as the divergence
between
the q distribution on external states
and the p distribution
on external states conditioned on
particular states
plus the self-surprisal of those
particular states or the one that's
closest to the bayesian information
criterion bayesian modeling world but
still being shown as equivalent to
is the accuracy minus model complexity
and the evidence lower bound
so that's where we we've seen these
formalisms come out
many many times
but we're approaching it from a very

different way and a lot more bottom up
and
i think but again would be open to
people
correcting
in a way that
fully embraces the map territory
distinction and makes no bones about it
whereas if you just pop in from the top
with these formalisms it might seem
like they're describing aspects of the
system but i hope that the way we've
approached this over the last like six
hours of live streams literally and the
way that the authors wrote it so
carefully makes it clear we're talking
about these formalisms on
probably not even the first map
and maybe the second map or summary
statistics thereof these are
relationships that are intrinsic to our
approach of map making
because we did the laplacian
approximation et cetera et cetera et
cetera
stephen d
yeah this is very helpful um i might
just ask we go
and walk through the very first sentence
again in a second just because i think
it'd be helpful just to reiterate but
from what you're saying and i i hear
this really clearly is
the the particular states
and and the thinness
of maintaining particular states
potentially or of whatever those
particular states

persist
effectively
then is
the same
when at a higher order level over time
and space as a policy
because let's not bring policy on
picture yet we just haven't gotten there
but particularly
okay i won't i won't jump the gun on
that but
um
i i suppose
the the the point is
just to say that
we we can still be in the realm of
thingness
okay we can still be in the realm of
thingness because we're talking about
particular states
of things
so we haven't got and i won't say beyond
that and then maybe just just to go
through the ontology not
just of that first sentence just to
reiterate just to get it clearer
again of autonomous flow
expected internal state could you just
recap that just to
just unpack that a little bit more just
so it's a little bit clearer
okay
so
the
autonomous flow
so autonomous states are actions
which are not we're not even bringing
policy selection in so it's the right

direction we want to go we want to be
talking about internal states that are
actually doing planning as inference
that affect their action states but for
now we're just treating the action
states as outgoing statistical
dependencies and then we're just we're
pausing on
how the actions are selected we're just
because in the lorenz system there's no
policy selection but there are external
states that are modified by states those
are the active states so it's almost
like we can talk about action without
policy inference
so the autonomous states
the autonomous flow
at the expected internal state
so these are expectations
of
autonomous states action
outgoing and internal states
can be expressed as a gradient flow so
now it's like
we're going to um rather than have a six
state system
we're going to
rewrite
as
four
remember we pulled out four kinds of
notes
and now we're going to express a flow
not in a six-dimensional space
where each dimension was a measurement
that we were making
but express a flow
over the four kinds of states

and then that means that even if there
was a bunch of internal nodes that were
communicating to each other
we could still use this four-fold
breakdown

william blake hashtag bucky hashtag
fristen

no matter how many specifics were inside
the internal

or how many blanket states there were
like how many nodes in the graph or how
many external states because now it's
like a flow over kinds
of nodes

and then that

because the nodes are defined by their
directionality of relationship

like how they influence each other right
like internal and external we know don't
directly influence each other they're
conditionally independent based upon
blanket states etc

that allows us to write formalisms that
are like evergreen

and can be expressed as flow

dean

yeah i would just say that when i when i
first encountered these kind of
formalisms

i always ask so and then what happened
and that's when i

coined the phrase when in doubt

zoom in zoom out but i didn't

want that zooming in and zooming out to
be collapsed just to the optical zoom

i assumed that your legs wouldn't become
paralyzed when you

picked up a telescope or a microscope

that you would still zoom in and zoom
out by
moving
right so the inference is in doubt
so how do you overcome that you become
more active but not just with your eyes
you actually pick yourself up and move
yourself around a little bit more
a la
you don't just have to be on on the
on the north end of the carousel
sometimes you can pick up a ladder and
look down on it as well and that will
give you a whole
new set of data whereas more
complementary data a different
collection
and so
that's why i think it's really really
good to be able to see that
and understand what it implies
it basically says if you're if you're
if you're not sure
act some more
how could i disagree with acting first
serve
let's just look at this slide and then
stephen so this functional
the flow over the four kinds of states
that is using
this blanket partitioning paradigm
can be expressed in several forms
expected energy
 e big e is expectation
expected energy minus the entropy of the
variational distribution remember the q
variational distributions like the one
that we get to control q of μ that's

the q of internal states
which is equivalent to
the self information
fancy i plus the KL divergence between
the variational and conditional density
that's this KL divergence term which is
always greater than zero so
self-surprisal is always zero or greater
you can't have negative surprisal
um
and then the divergence is always going
to be positive as well
this can be decomposed into
accuracy and complexity
so that's this third framing and in that
setting
negative free energy becomes the
evidence lower bound or elbow which will
be familiar to many people with machine
learning backgrounds
okay amazing so those are the three
different formalisms and how they fell
out
this is the basis of the free energy
principle wait what's the free energy
principle
put simply it means that the expected
internal states of a particular
partition
at non-equilibrium steady state
can be cast as encoding conditional or
bayesian beliefs about external states
that's the free energy principle
in this paper
not the first not the last way it'll be
defined and not the only but that's how
they
discuss it here

and basically we moved from early on
it's like tying our shoe with a density
to dynamics side
and now
we have this whole universe of
formalisms
that connect
free energy-looking formalisms like
chemistry
with information theory formalisms
so
thermochemistry with energy and entropy
with
thermal information
divergence and self-information
to
information and inference with accuracy
minus complexity
as
a modeling heuristic
so that's big
free energy is big
and
then they immediately go to talk about
physiological perspectives
and how
that surprisal function
that constitutes the potential
is log model evidence so
if you're getting good evidence for your
physiological model you're spending a
lot of times in your preferred and
expected physiological states
and that can be seen as simply a
statement of homeostasis
where outgoing statistical dependencies
i.e active states
maintain interoceptive signals

i.e

incoming state sensory states within
characteristic physiological ranges
you

minimize your surprise about that
until you don't and you dissipate
stephen

amazing

this this really ties into the active
infant's paradigm shift

because okay

traditionally if we're going to have
represented like traditionally
representationalism for instance

like you were saying it does really
really

throw that out in a way because

if i have 100 cameras normally

and i took 100 different cameras all

taking a photo it's going to make it 100
times more

blurry in in that sense however in this
sense

a hundred sources

in this

noisy

dynamical

gives me a hundred ways

to

integrate

free energy reduction

and even actually the noise of me moving

it 100 ways by 100 cameras gives me more

and more informational

thermodynamic

ways in

so that that just shifts very much the
way

we might think about how

to know about

how to act in the world

cool

um

let's continue on and we'll probably do

a few minutes after the hour to get to

these last few points

so

we've been talking about

different ways that we can talk about

the free energy functional of

the particular states

5.1

is where

this idea of the internal states as a

generative model and self-evidencing or

like the minimization of self-surprise

is really leaned into so the alternative

and deflationary perspective

rests on noting that the free energy

gradients are also the gradients of

self-information so it's like the free

energy itself

is a wind that's difficult to latch on

to so the free energy bowl the one where

you'd be doing strictly as well as you

could if you were on the bottom of it

we're not on that bowl

but the one that we can

be keenly aware of is our self-supprisal

so if

the self-supprisal gradient

is aligned with

the free energy gradients

then

reducing our uncertainty

about sensory states incoming

helps us

on the way towards minimizing free
energy gradients reducing that
difference

the organism wants to be minimally
surprised about expected and preferred
observations

and those are the perceptions or
measurements

more technically it wants to be
minimally surprised about expected
states

and then preferences are going to play a
key role when we talk about active
systems that do resist dissipation
this is where the moth to the flame
comes and i just i love this plot arc
from 2006

ice queen fristen with the winged
snowflake

and then you know things are heating up
for act dimp

others are being drawn like a chaotic
moth to the flame they come close but
then it burns and then they fly away
so one could imagine these kinds of
systems to be like stochastic chaos
and that could be now again we're
thinking about previously the lorenz
system was like maybe

a particle in a chaotic weather system
so that was like the weather systems
that lorenz was studying

but now we're going to be thinking about
particular states like a particular moth
that's able to

not just have outgoing statistical
dependencies

but to do
policy selection
so this is where we actually get
adaptive active inference and again you
know if there's two kinds of moths some
of them are just particles and then they
get burnt when they get burnt and they
don't when they don't
some will survive and some won't but the
ones that do
enact
effective policy selection
are going to be persistent even a day
later so then the lap adopter have been
around for millions of years so is it
really any surprise that the moths have
adaptive
behavioral systems
but that when the niche changes like
with electric nighttime lighting that
that doesn't entail that it's going to
instantly be the adaptive system for
that new niche
we can think about the motion of a moth
attracted towards a flame
but constantly thwarted by turbulent or
solenoidal air currents
because active states influence sensory
states and possibly external states this
would look as if the particle the moth
was trying to attain its most likely
state in the face of random fluctuations
and solenoidal dynamics
so
it's like there's those arrows the
irrotational component are like
converging the moth to the flame
but then like overshoot and the windy

room are blowing it around and so again
there's the moth in the room and we
could look at it in the x y and the z
and if the candle were in the center of
the room
it would be like a bowl
that the moth were converging towards
for any two dimensions that you looked
at and indeed all three dimensions when
considered together
and the movement of that particle the
dynamics of that particle
would be related to something about the
flow of the air in that room
with a pullback attractor
this perspective emphasizes the active
part of self-evidencing sometimes
referred to as active inference so we
were calculating the potential the
self-surprisal the outgoing dependencies
for particles before we talked about
policy selection
and that is showing mere
active inference as it's called but then
when you think about the real world
where only the anti-dissipative systems
persist and the successful ones even
more so
that's where policy selection comes into
play
and that's actually where this paper
ends
they basically just
bring it to that point
and say all systems are going to be able
to be framed within this way again not
the territories
all of the approximations that we make

will have these attributes
and then
we can imagine that there are some
systems where the internal states are
not just acting as if they're doing
conditional expectations on external
states but then they take those
conditional expectations like about the
weather
and then there's another node for policy
selection like the other pi
policy selection in the models we've
seen
and then that pie can be like put on the
jacket or not those are like the two
affordances
or those are the two states in the
affordance variable
and then
in a dissipative world
you're going to see
adaptive agents as the ones that are
able to recurrently adapt to changing
and chaotic circumstance to repeatedly
come back to that candle
and so just to summarize then we'll just
um
if there's any implications we want to
touch but we'll summarize
um they constructed a laplacian system
that's quadratic aka bowl like
in its potential function and flow
operators state dependent flow operators
by the way that includes the
housekeeping function
that evinces so provides evidence for
stochastic chaos because it has that
leopardive exponent that was very

similar to the ones that were calculated
from the non-laplace assumptions or
approximations
the way that the polynomial
approximation played out with the first
and second order approximations
gave a sparsity
which
can be interpreted as conditional
independencies
and a partitioning into four kinds of
states
coupled systems
have
a conditional synchronization map
in this example it had a linear form
they were on that manifold that was like
 $y = x$
and this is important because in these
laplacian coupled systems
the conditional synchronization map is
necessarily diffeomorphic again would be
awesome if somebody wanted to come on
and like help us see what the technical
implications of the diffeomorphism is
but
the simplest
level is like it doesn't bend back on
itself
so it's a good function
so that means for every sensory state
there's a unique expected internal state
and conditional density over external
states
so it'd be like yeah when the
thermometer says like 20 it could be 20
or 50.
that's not a very helpful thermometer

now if you say the thermometer always
reads half of the temperature or it's
always double or it's a little noisy
those are all workable but like the
thermometer could just totally be
deceiving you and then it just is a it's
a wacky world with what the thermometer
says

not going to be a very adaptive system
the resulting formulation can be read as
bayesian mechanics

in which internal states on average
parameterize belief of a bayesian sort
about external states

just like these equations

when applied to particles and

statistical distributions were

statistical mechanics

we're taking the full bayesian insight

and reading these equivalencies and

formalisms as a bayesian mechanics

so that's the summary of 5-2

and then there's a few implications that

they get into but we'll

first have any questions so dean and

then

stephen well i just think it's

fascinating that

it doesn't matter whether we're talking

moths or or people we're kind of

blissfully unaware of the statistical

uh manifold

floating around us prior to us being

pulled back to that flame and either

becoming the hero moth

or the fool moth we don't discover that

until afterwards

but if we could be aware of the kind of

the statistical
distribution in play prior to us
being pulled it might help
i don't think it don't i don't think it
determines much
but it certainly
enlightens as opposed to being lit up
yeah in the moth example i think well
you know but if it really reaches the
flame it dies so maybe that's not even
the final perfect example for how
attractors work that'd be like a bowl
with a hole at the bottom so it again it
just it's it's a heuristic it helps us
think about the kinds of systems that
we're talking about but
um realism interpretation asides even
that example doesn't capture the whole
thing because it's a metaphor because
it's like map of map of territory so
it's totally chill stephen
yeah this uh
the evincing or even says stochastic
chaos which
wasn't a word i'd heard of before to be
honest um is this idea of equality um
so that that's really
useful that how
it gives a sense of how
um
both there can be
chaos in the attractor and the noise
maintained at least at some level
throughout because the
um in most cases the noise is just um
averaged away in in other scenarios so
this is
this is this this piece where we tapping

into that um

quality

is really interesting and then that then

tying into the bayesian mechanics all

the way down

um is uh

very very useful especially when i think

the bayesian mechanics as i mentioned

before

is stacked it's like

it's not saying

this is where everything is different to

evidence in it's like like we're going

to give better evidence which or better

evidence which make us think we're going

to show a more accurate lorenz attractor

solution no it's

evinces

and we can stack up and add these fuzzy

qualities together

and we'll still have a fuzzy quality but

maybe we'll have one which is a little

bit

more

practical and that's the game in town

here is practicality so i like that who

also um

it does average out noise

the internal states on average

the expectation

parameterizes beliefs and so that's

going to be a major challenge

is to go from these expectation and mean

field approximations

without going into ergodicity that is

literally the whole challenge

the laplacian is like a quadratic that's

fit

on another distribution which we're not
even caring to learn the form of
and so we could say that the expectation
the single number expectation the
expectation of a child in that class is
five feet tall

but maybe it's all four and six feet but
if we fit a laplacian on that
it's gonna be at five looking like a
quadratic

so the map and the territory are not the
same

and it is a significant challenge across
domains to connect some of the
approximations

that

are double-edged swords

like let's approximate over a bunch of
possible futures

or let's approximate this way like
there's no

free lunch

so the internal states on average
parameterized beliefs but does any
given

internal state parameterize beliefs

evolution kind of helps us out by saying

like well on average it's going to get

better and better that's like the

fundamental theorem of natural selection
ftns

and then the ones that don't do as well
they die

so evolution in a way helps us get out
of this

but the formalism doesn't have that as
we saw it right now yeah stephen on that
that's a really good point and and so it

also does tie into the
so it gives away to sort of the word
average but yeah to average or
approximate these faster perturbations
yet it still maintains perturbations
which could be used at the next nested
level or in other ways so it's so
there's an interesting
question there's an interesting thing
about
yeah it's reducing and
um
the the
the
overwhelming noise
yes it's also
not doing what traditional statistical
averaging does which is basically crash
everything down to
you know a fixed point
um it gives this ability for it still to
be alive in a way
absolutely like if we just took the
expectation the single average point
of
the lorenz attractor itself
it would just be one point in the middle
of the saddle
okay
[Music]
but we have something that still has a
dynamical nature
with a tractable approximation
and the dynamics have a positively open
of exponent
so we didn't squash the system down to a
point or a racetrack
we kept

some of the dynamical and even chaotic
aspects of the system
while still being able to talk about
things on average
but we're not talking about the average
position of the particle
yes even quickly on that
yeah i think that's a good point and
also just one thing that caught my mind
is with that butterfly with the moth
going to the flame
is it talks about a perspective that's
been taken
yet
and this is probably useful when people
are thinking
in the applied active inference this is
the perspective that we're having and
it's possibly more used the word
perspective because we see a picture of
a moth if there was no picture there we
might say a description
of the scenario okay so there's this
perspective
on the moth
yet the moth isn't necessarily in this
scenario taking a perspective it's
doing
it's it's doing its mothness
yep it's definitely moth's gonna moth
although the picture was not in the
paper i added that one but yeah the
winged snowflake was indeed in the 2006
paper so let's just recap a few of the
domains of implication and then we'll
close because like
what a fun
paper and discussion

so
one implication
um but also sort of
note of caution
is that just because we took the original
complex system lorenz and coupled it so
two
chess moves beyond some sort of simple
scenario
it doesn't mean that
a synchronization map exists
let alone a good one for any given
system but
one of the implications of this paper
was that now we know that for a non-zero
subset of dynamical systems
a non-zero subset of chaotic dynamical
systems and a non-zero subset of
dynamical chaotic coupled systems
we can get this well-behaved
approximation map
so not to say all but we know that the
answer is now
not none
so that's a key implication
another implication was to really start
to bring in the variational free energy
in terms of generalized synchrony that's
what opens up to the de costa discrete
state space synthesis active inference
bayesian surprise
optimal bayesian design intrinsic
motivation info masks
risk sensitive policies kl control
occam's principle decision theory
utility theory max ends constructal
theory like
this is not all

in this paper so a lot more has to be
done to get to some of these places
especially in such a
reasoned and bottom-up way
another implication of the paper is that
took the helmholtz decomposition in a
new way
and it might provide a genetic generic
model for random dynamical systems
importantly because we connected it so
closely to bayesian
generative modeling
it could be possible to do less just
like
let's do descriptive analytics on flow
to like
let's have a map
that we can run both ways
where we're updating the parameters of
our flow map
with what we're observing but then we
can also generate data sets we can
generate counter factuals and so that is
what allows sparse data sets to play
into active and fep
and ties us more closely to modern
machine learning like variational auto
encoders
another implication
um or
next step
would be like we were looking at a
three-dimensional system the x y and z
of the lorenz attractor
but
how could we approach high dimensional
dynamical systems
so attractors in 50 dimensions that's a

little bit different and hard to solve

so they suggest that a way to approach

this

probably hint cough cough wink wink etc

is to learn

the state dependent flow operator

and the nests

using deep neural networks

so it's kind of like

you have a neural network now that

recognizes bulls

and so we're still kind of keeping some

of the same things that we had a

intuitive interpretation of in lower

dimensions

it's just like you go like okay i know

about the cube and then someone says

well there's like an 11 dimensional cube

and you okay

i kind of know what i don't really see

exactly what that means but i kind of

see what is being discussed how are we

going to compute on the 11-dimensional

cube though well neural networks

and that is actually very close to

neural stochastic differential equations

and i think this is super exciting

one could ergo hasn't been done probably

hint hint

use a separate feed forward neural

network

to parameterize different components of

the flow operator and self-information

so in other words we're taking what we

talked about here

and we can think about combining it with

the types of stochastic differential

equations neural stochastic differential

equations and bayesian neural networks
all of these uh advances in machine
learning and now we're actually
coming back to the table
with new architectures for deep learning
because this would be a deep learning
model it would have a deep neural
network you know you would benefit from
having a gpu
but it's an architecture that was
inspired by the discussions that we had
here
and then just the final implications and
then we'll just have a little closing
um
the flows across the boundary
also notice that a little bit of the
blanket starting to potentially be
inched away from talking more about a
markov boundary
and connecting the fep
to the constructal law
and
the work of professor adrian bakan who
honestly is a lot like fristen in some
ways and a really gracious person and
communicator as well with a long career
of working towards this kind of a
unifying flow theory to get towards the
evolution of design in nature and as the
image shows like kind of like from flood
plains to lung alveoli that similar kind
of a unifying model
and then
a final
set of implications
is that we take covariances
all the time from time series

that's like how we do
crypto trading that's how we do weather
prediction that's all kinds of systems
take covariance matrices on time series
and so potentially
because
the hessian is recapitulating elements
of the covariance matrix
it could be possible to have an
empirical data set
and empirically take the covariance and
then learn something about the hessian
the curvature of the underlying flow
states so in other words if you use the
laplacian approximation even if it's
stochastic chaotic and dynamic
that empirical covariance could identify
conditional independencies where there's
a zero
and so in principle
you could furnish a description of
expected flow and the leopard of
exponents to establish whether the
system was chaotic or not because we
found that the hessian
approximation
gave us a leopard if estimate that was
like the actual leopardive exponent in
the lorenz system so we could go from
empirical covariances which people do
literally every day cov open parentheses
and r
empirical covariances
two
estimated
hessians two
estimates of things like the lyapuniv
exponents

and that's why people often study
chaotic systems in the super toy models
like the double pendulum and lorenz
because it turns out it's shockingly
hard to find chaotic systems in nature
outside of those toy examples because
the real noise in the system swamps your
signal
and so in conclusion
having a simple functional form for the
flow of random dynamical systems may be
useful for modeling and analyzing time
series
generated by real-world processes that
are far from equilibrium and may or may
not be chaotic and so that's the closing
question
human flow
physical flow supply chains information
flow narrative flow semantic flow
ants bringing seeds in and taking the
mid and out all of these flows
we have data sets for them
and now we have a new kind of model that
we can use to model those flows
so we'll just chill and talk for a few
more minutes but what a great discussion
so thanks a lot dean and steven and
everybody else who participated in 32.
so stephen first
deep breath
yeah i i i there was a couple of things
there that seems to have a big impact um
because
as well as markovian monism and the idea
that active inference brings in this um
this this this partitioning and the the
active inference field

um which can then also have
relationships to deep learning and how
predictive processes thought about
what's being said here that there's also
evincing or evidence the idea of the
evidence and statistic chaos
itself also been a paradigm
a paradigmatic
uh
part of
evidence in being arrived being being
arrived at
so
which in some ways is not dependent
purely on markovian monism being
the the the overarching game in town
so that is quite a um
that's quite an interesting additional
way for
deep
learning
to
be um
to be impacted by
is this a field of practice is it a
domain i don't know what we call it but
the that that does
it does it does make everything
does it doesn't mean it's all markovian
monism
in terms of the ability to then infer
you've also got this other
um ability to
get
information and from what you're saying
it almost as much by inferring if there
is chaotic
presence or if there's not maybe not

having maybe going and seeing if there's
not
chaos present
may give an idea of there's maybe no
correlations present and that's that's
to jump in the gun but you know maybe
there's other ways of being able to
interrogate statistical dependencies in
time series analysis
that can be sitting alongside
the markovian monism
so how that's connected to mycovian
monism is definitely a good question but
yes
these are general tools for complex
systems analysis so
i hope people don't read this paper as
like fristen apologetics or some sort of
fringe development
in act imp only it's pretty clear from
the nouns that we used that this is
actually like a fundamental contribution
towards analyzing dynamical stochastic
chaotic systems however just because
you find for example a lowly operative
exponent of your approximation
it's not a statement about the system
so we will always be able to just
halt right there you know it's not that
it's not a chaotic system
your approximation
doesn't have a chaotic leading lyapunov
exponent
so say what you want to say about your
map now if your map isn't chaotic
and you're doing a great job
predicting explaining controlling
designing creating

wonderful i hope the business is
successful and that you can you know
have friendship and health and all those
things but that's not a claim about the
territory and so there's going to be so
many fun ways to talk and explore that
dean

what are your closing thoughts

just real quick i think

what the co-variation piece which is
basically statistics and an abstraction
to to sort of explain things

points out is that

if we can understand

the world as being both within

enveloped

and between partitioned at the same time

that's a good fundamental place to start

and i completely agree with your last

statement we're not

we're not counting fairies on pins

and then just leaving it there we're

actually hoping at some point to be able

to turn this into something useful and

functional but if you can't if you can't

start from a place of seeing it as two

things at once within in between

you're going to struggle

you're going to tend to collapse

prematurely and you're not going to get

very creative and active inference will

well you'll be able to define it and

describe it but i'm not sure how how

much

new information you'll be able to derive

with it unless you can hold up both at

once it's within in between

great there's so much more that we

all have to learn the appendices the
formalisms we didn't go into the
citations
other perspectives
you know it's just the beginning and
that's what dot two is about like
how far we've come you know when we can
go back to our road map and just be like
wait dynamical systems and densities and
then we're you know
300 miles past there but
where have we gotten
and so it's like
just really exciting and i'm glad that
we're
doing it in a participatory way because
there's a lot to it so
dean and steven thank you again
everybody watching live and in replay
hope that you get involved and stay
involved with actin flap see you later
bye
you